EXPERIMENT NUMBER: 7

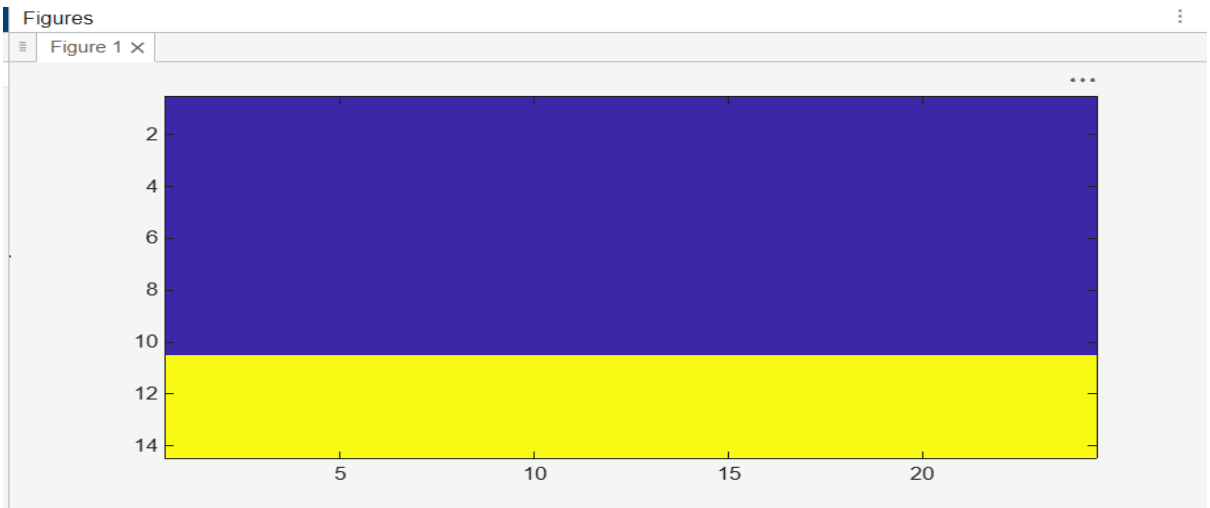
Analysis of 5G NR Frame Structure and Subcarrier Spacing

OBJECTIVE 1:

Matlab Program

```
clc; clear; close all;
Nsc = 24; % Simplified number of subcarriers
Nsym = 14; % OFDM symbols per slot
grid = zeros(Nsym,Nsc);
% Simplified DL assignment: symbols 1-10
grid(1:10,:) = 1;
% Simplified UL assignment: symbols 11-14
grid(11:14,:) = 2;
figure
imagesc(grid)
gridon;
xlabel('Subcarrier Index')
ylabel('OFDM Symbol Index')
title('Simplified 5G NR Time-Frequency Resource Grid')
colorbar
```

Result

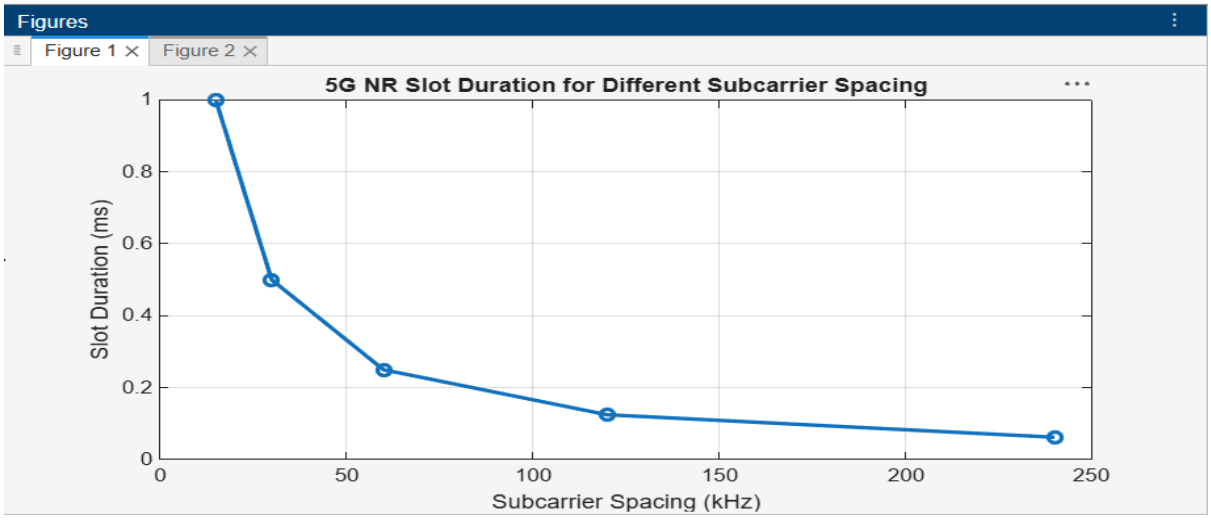


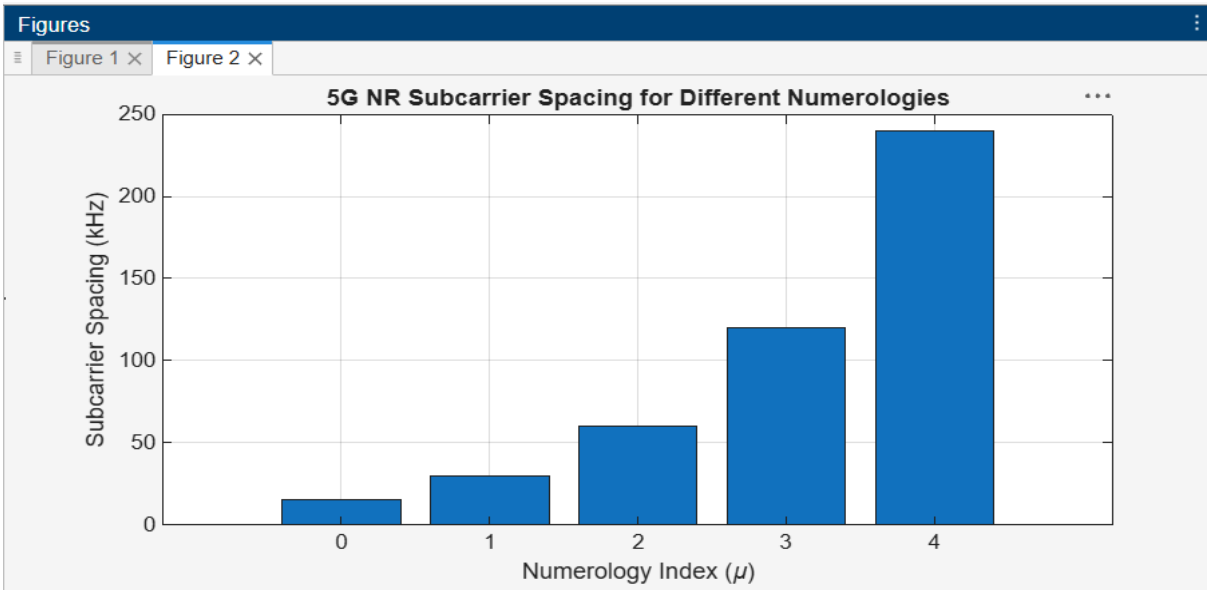
OBJECTIVE 2:

Matlab Program

```
clc; clear; close all;
scs = [15 30 60 120 240]; % kHz
mu = 0:4;
slot = [1 0.5 0.25 0.125 0.0625]; % ms
figure
plot(scs,slot,'o-','LineWidth',2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
figure
bar(scs)
grid on
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

Result



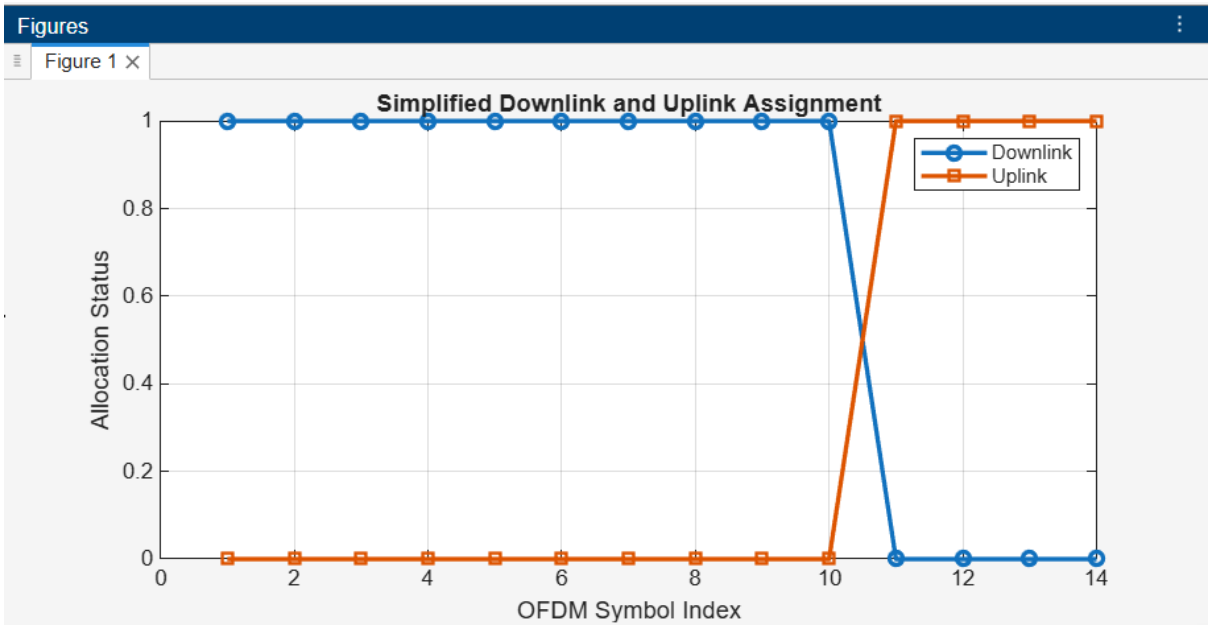


OBJECTIVE 3:

Matlab Program

```
clc; clear; close all;
symbols = 1:14;
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];
figure
plot(symbols,DL,'o-','LineWidth',2); hold on
plot(symbols,UL,'s-','LineWidth',2)
grid on
xlabel('OFDM Symbol Index')
ylabel('Allocation Status')
title('Simplified Downlink and Uplink Assignment')
legend('Downlink','Uplink')
```

Result


Figures
Figure 1
Simplified Downlink and Uplink Assignment
Downlink
Uplink
Allocation Status
OFDM Symbol Index
0
0.2
0.4
0.6
0.8
1
0
2
4
6
8
10
12
14